

Disaster Education Media Competition
USK GLOBAL AWARD ON DISASTER RESILIENCE 3.0

**Escape Disaster: Game-Based Media for Early Disaster
Preparedness Education**



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**Escape Disaster: Game-Based Media for Early Disaster
Preparedness Education
Syiah Kuala University**

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Summary

Indonesia sits on three major tectonic plates in the Pacific Ring of Fire, which makes it vulnerable to disasters such as earthquakes, eruptions, and tsunamis. This shows the urgent need for disaster preparedness education, especially for children who will inherit the future. In the effort of early education regarding disaster preparedness, particularly in providing understanding of self-rescue steps, the Escape Disaster game emerged as a medium to teach children about disaster preparedness. Escape Disaster is a mobile-based digital game with a quiz style gameplay designed to teach safety measures during disasters. The game aims to educate and train children on how to prepare and respond appropriately when facing a disaster. By using mobile games as a medium, it is expected that children will not only play but also gain disaster knowledge and be prepared with self-rescue skills from an early age. Designed with an engaging and enjoyable gameplay, Escape Disaster ensures that learning about disaster preparedness remains fun and interactive. Through Escape Disaster, children will be able to make the right decisions when a disaster occurs.

Keyword : Escape Disaster, Disaster Preparedness Education, Mobile Games, Children, Safety Measures

Chapter I – Introduction

a. Background

Indonesia lies at the confluence of three major tectonic plates: the Indo-Australian, Pacific, and Eurasian Plates which is within the Pacific Ring of Fire, one of the most seismically active regions in the world (De Paor, 2017). This position creates more than 70 active faults and numerous subduction zones, exposing the country to frequent earthquakes, tsunamis, volcanic eruptions, floods, and landslides (Hinga, 2015). Additionally, the presence of the Alpide Belt, the second most active seismic zone globally, further intensifies Indonesia's vulnerability to natural disasters. These conditions highlight the urgent need for disaster preparedness, not only as theoretical knowledge but also as practical skills for survival and response in real-life emergencies.

In recent years, disaster education has shifted from a focus on merely transferring information to emphasizing the development of critical skills and resilience (Guo et al., 2025). Education is no longer expected to only build awareness but also foster the ability to act effectively during crises. Elementary school students represent an essential target group for such education, as they are in a formative stage of character and knowledge development. Instilling preparedness at an early age can reduce panic and encourage calm, rational decision-making in emergency situations (Wardana et al., 2021).

Traditional disaster education, however, often remains theoretical and monotonous, limiting student's engagement and comprehension. The lack of interactive media has made it difficult for children to fully internalize disaster preparedness concepts, which require both cognitive and experiential learning. Without engaging delivery methods, disaster knowledge risks becoming abstract and ineffective in practice.

To address these challenges, the use of interactive, technology-based media has emerged as an effective strategy. Educational games and simulations provide immersive learning experiences that capture attention, stimulate curiosity, and cultivate decision-making skills. Mobile-based platforms, in particular, align with the digital habits of today's children, offering an accessible and enjoyable medium to promote disaster awareness (Salma et al., 2025).

Responding to this need, the Escape Disaster game was developed as a mobile-based educational tool that combines entertainment with simulation-based learning. Designed for elementary students, it integrates disaster scenarios with interactive quizzes that train children to think critically and make quick, informed decisions. By leveraging digital engagement, Escape Disaster aims to increase disaster preparedness, enhance response skills, and foster resilience in the younger generation through a fun and interactive approach.

b. Problem Formulation

This problem statement arises from the low level of understanding and awareness of disaster preparedness among elementary school children (Herdiansyah et al., 2020). The existing disaster education process tends to be theoretical and monotonous, which makes it unappealing for children to learn. On the other hand, children are more familiar with digital technology and interactive mobile games (Gocheva et al., 2020). This situation raises the question of how to design a mobile game that not only provides entertainment but also effectively incorporates disaster education. Furthermore, it is necessary to examine how the game mechanics can help children practice making quick decisions when facing emergency situations, and to what extent this game can contribute to increasing disaster preparedness from an early age.

c. Purposes

The objective of developing the Escape Disaster game is to create a mobile-based educational medium that can instill disaster awareness and preparedness from an early age in elementary school children. Through interactive gameplay, especially the decision-making simulation quiz feature, the game is expected to provide a fun learning experience while also increasing children's knowledge about the steps to take when facing a disaster. Furthermore, this game aims to improve children's critical thinking skills, quick decision-making, and foster an emergency response attitude that will be beneficial in everyday life.

d. Gameplay

The benefits of implementing Escape Disaster can be summarized as follows:

1. It trains critical thinking and quick decision-making skills for dealing with disaster situations.
2. It contributes to improving education on how to survive and save oneself during a disaster.
3. It provides an interactive learning medium through disaster simulations and quizzes tailored to specific needs and types of disasters.
4. It offers educational entertainment that can attract public interest, especially among the younger generation, to learn about disaster preparedness in a fun way.

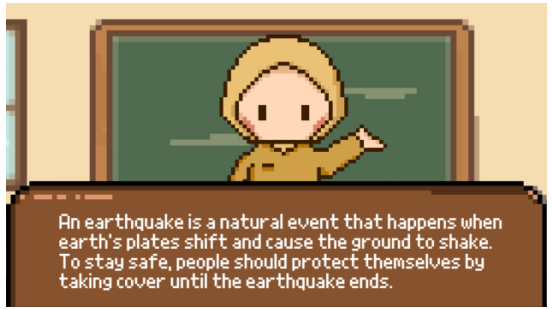
Chapter II – Design and Concept

a. Game Design

The Escape Disaster game is designed as a mobile-based educational tool that combines entertainment and educations about disaster preparedness. The apps we used to design this escape disaster game were ibis Paint and Canva. The game's design is simple, interactive, and easy to understand for elementary school children, the primary target audience. The user interface uses bright colors, imaginative icons, and child-friendly animated illustrations to create a fun learning experience.

The game's interface consists of:

No	Game Design Figure	Description
1		The logo of the Escape Disaster mobile application features a character illustration accompanied by wave imagery, symbolizing the theme of natural disasters. The game title Escape Disaster is prominently displayed within the design to ensure clear identification. This logo serves both as a visual identity and as a representation of the game's core focus on disaster preparedness education

2		The Start button in Escape Disaster is prominently positioned at the center of the opening screen, directly below the game title. It is designed in a rounded rectangular shape with contrasting colors to enhance visibility and draw user attention. Functionally, the button provides an intuitive entry point for players to begin the simulation. Its simple and accessible design reflects usability principles, making it appropriate for the target audience, particularly children.
3		At the beginning of the Escape Disaster game, a simulation scene is displayed in which a teacher provides disaster education to students. This segment appears before the gameplay challenges begin, serving as an introductory learning phase. The visual emphasizes the educational purpose of the game by presenting disaster preparedness education in a comprehensible and engaging way.
4		Following the educational simulation, the Escape Disaster game presents a quiz challenge as a measure of players' understanding on disaster preparedness material. The quiz is designed in an interactive format, allowing players to select appropriate responses to given scenarios. To enhance engagement, a countdown timer is provided as a time limit for answering, encouraging children to practice accurate and quick decision-making. This stage functions as both an assessment tool and a reinforcement of the educational content delivered at the beginning of the game.
5		A “green check mark” is displayed as positive feedback when players answer the quiz correctly and within the specified time. This visual symbol functions not only as an appreciation of accuracy but also as confirmation that the player may advance to the next stage of the simulation, thereby maintaining motivation and engagement in the learning process. When players provide an incorrect response/ fail to answer within the time limit, the game displays a “Game Over” screen. This consequence highlights the importance of accuracy and <i>urgency</i> in decision making, reinforcing the educational purpose of the game by illustrating the risks of unpreparedness in disaster situations.

6		After completing the self-rescue quiz in the classroom setting, players advance to the next stage of the Escape Disaster game. At this level, the scope of the simulation expands beyond the classroom environment, transitioning into outdoor disaster rescue scenarios. This progression is designed to provide a more comprehensive learning experience by exposing players to varied contexts of disaster preparedness.
7		After moving beyond the classroom environment, players are introduced to an outdoor disaster simulation illustrated through a tsunami scenario. At this stage, the game provides a brief disaster education session followed by another quiz, enabling players to apply their knowledge and practice decision-making in a more realistic context.
8		The settings menu in the Escape Disaster game allows players to adjust music and sound effect volumes, choose between English and Indonesian language options, and return to the main menu through the home button. This feature ensures a more comfortable and accessible gameplay experience.

Figure 2

Design Table : This table shows and describe each of Escape Disaster game design.

b. Game Concept

The main concept of the Escape Disaster game is a decision-making simulation during a disaster. Escape Disaster is a game that falls into the category of educational games. The game progresses through levels that act both as stages of difficulty and as markers of how well players grasp survival strategies. Each level introduces key disaster-related information to ensure players are ready before the simulation begins.

Players will be faced with various self-rescue scenarios during disasters, such as earthquakes, floods, or fires. They must answer interactive quiz questions that represent different actions to take in an emergency. Every answer has a different

consequence, which helps children learn which actions are appropriate for self-rescue. Each quiz question has a time limit, forcing players to make quick decisions. Based on the speed and accuracy of their answers, children will be trained to think critically, make quick decisions, and understand safety procedures.

c. Main Feature

1. Disaster Knowledge, Materials about different disasters and how to act safely during them
2. Disaster Scenarios, Various disaster situations presented to show what can happen in an actual disaster.
3. Interactive Quizzes, multiple-choice questions that test players' decisions in emergency situations.
4. Level System, the higher the level, the more complex the disaster challenges faced.
5. Educational Visuals, simple animations and illustrations that guide children in learning safety steps.

d. Gameplay

The gameplay begins with the player entering a simulation in the form of a short story that illustrates the disaster situation. The player is then given several action choices through an interactive quiz. If the answer is correct, the player can unlock the next level. If it is wrong, the game will stop, and the player must start over. The game ends after all challenges are completed and can then continue to the next stage.

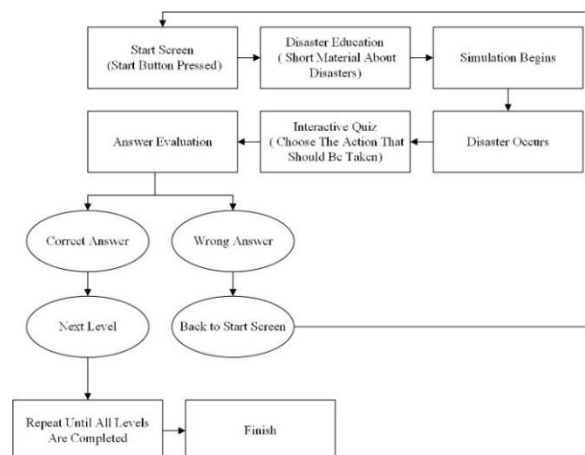


Figure 2

Gameplay Flowchart : This diagram illustrates the flow of the gameplay process.

Chapter III – Implementation Methodology

The implementation methodology for the game Escape Disaster is structured systematically to ensure the development process is well-directed and aligned with its disaster education goals. The stages are as follows:

a. Needs Analysis

1. Identify the problems in disaster education, especially for elementary school children who are often less interested in conventional learning methods.
2. Gather data from literature, BNPB disaster modules, and disaster mitigation guides to ensure the educational content meets the required standards.

b. Concept Design

1. Determine the game mechanics, storyline, materials, disaster scenarios (earthquakes, floods, fires), and interactive quiz features as the core of the learning process
2. Design characters, avatars, and environmental illustrations with a child-friendly visual style.

c. Interface and Visual Design

1. Create the main menu, settings, disaster scenarios, quiz screens, and action illustrations.
2. Integrate educational elements in the form of simple materials on self-rescue steps.
3. Implement the design into a mobile platform using the TurboWarp game engine.

d. Game Development

1. Add more interactivity, difficulty levels, and disaster animations.
2. Include more types of disasters, such as landslides, volcanic eruptions, and tornadoes.

e. Testing and Evaluation

1. Conduct limited trials with elementary school children to assess their level of engagement, material comprehension, and quiz effectiveness.
2. Gather feedback from the trials to improve the final version.

f. Implementation and Socialization

1. Integrate the game with disaster simulation activities as teaching material so children become more accustomed to real-life situations.
2. Socializing the game to parents so that they understand its educational value.
In addition, distributing the game through mobile platforms will also expand access for the general public.

g. Project Timeline Table

No	Stages	July				August				September			
		I	II	III	IV	I	II	III	IV	I	II	III	IV
1	Needs Analysis												
2	Concept Design												
3	Interface and Visual Design												
4	Game Development												
5	Writing Paper												
6	Testing and Evaluation												
7	Implementation and Socialization												
8	Paper Submission												
9	Further Game Development												

Figure 3

Project Timeline Table, Project phases and their durations are summarized in this table.

Chapter IV - Conclusion

Escape Disaster is a mobile-based educational media innovation designed to instill disaster awareness and preparedness from an early age, especially for elementary school children. Through the concept of interactive gameplay combined with disaster scenarios and decision-making quiz challenges, this game provides a fun and engaging learning experience, avoids monotony, and at the same time builds a fundamental understanding of self-rescue steps during a disaster.

With its simple design, child-friendly visual, and level-based system, *Escape Disaster* provides an engaging medium for disaster education. Which not only entertaining but also supports disaster learning by being adaptable in both classroom and indenpendent setting, keeping it enjoyable and educational for kids.

With the presence of Escape Disaster, disaster education is expected to be more easily received, understood, and instilled from an early age, so that the younger generation will have a better mental readiness and knowledge in facing any potential disasters in the future.

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I hereby declare that the work I have submitted for **The Disaster Education Media Competition (DEMC) – U-DARE 3.0** is entirely my own creation, not a plagiarism of someone else's work, and has never been entered into any competition or published anywhere.

If it is later discovered that my work does not comply with this statement, my submission will be automatically disqualified. This statement is made truthfully.

Banda Aceh, 30th August 2025

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